

Q2 Project Presentation:

Locally Adaptive k-Nearest Neighbors for Heart Disease Classification

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Problem

- Heart disease is a leading cause of death
- ML models can help diagnose them
- Medical ML needs to be interpretable and accurate

Why KNN is limited

- Standard KNN uses a single global k -value
- This assumes that the data is uniformly distributed
- Real medical data is mostly diverse

Research Question

Can we improve KNN by allowing the neighborhood size to adapt to local data density?

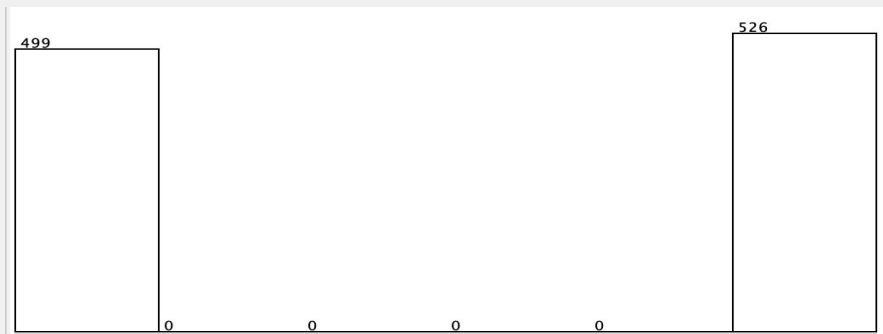
Our Proposed Method: LA-KNN

- Locally adaptive k-Nearest Neighbors
- Computes a different k value for each test point
- Based on local density estimation

$$k(\mathbf{x}) = \left\lfloor \frac{\alpha}{d_m(\mathbf{x}) + \epsilon} \right\rfloor$$

Dataset

- Heart Disease Dataset from Kaggle
- 1025 Instances, 13 features
- Binary Classification
- 80/20 Split



Data Header

age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
52	1	0	125	212	0	1	168	0	1	2	2	3	0
53	1	0	140	203	1	0	155	1	3.1	0	0	3	0
70	1	0	145	174	0	1	125	1	2.6	0	0	3	0
61	1	0	148	203	0	1	161	0	0	2	1	3	0
62	0	0	138	294	1	1	106	0	1.9	1	3	2	0
58	0	0	100	248	0	0	122	0	1	1	0	2	1
58	1	0	114	318	0	2	140	0	4.4	0	3	1	0
55	1	0	160	289	0	0	145	1	0.8	1	1	3	0

Preprocessing

- Median imputation
- One-hot encoding
- Z-score normalization

Noisy Dataset Creation

- 10% feature corruption
- 8% flipped labels
- Random feature scaling (0.5x - 2x)

Standard KNN

- Fixed k value
- Distance-weighted voting
- Tested k from {1, 3, 5, ... , 31}

LA-KNN

- Density based equation
- Alpha from {0.5, 1.0, 2.0, 4.0, 8.0}
- M from {3, 5, 7}
- ϵ to prevent dividing by zero

$$k(\mathbf{x}) = \left\lfloor \frac{\alpha}{d_m(\mathbf{x}) + \epsilon} \right\rfloor$$

Experimental Setup

- Compared 9 KNN models and 15 LA-KNN models
- Metrics:
 - Accuracy
 - Macro F1
 - Confusion Matrices

Results: Clean Dataset

- 100% Accuracy for both models
 - One classifier predicted label too well

Table 1: Performance Comparison on Clean Dataset

Method	Parameters	Accuracy	Macro F1
Standard KNN	$k = 1$	100%	1.000
LA-KNN	$m = 3, \alpha = 0.5$	100%	1.000

Table 2: Confusion Matrix - Clean Dataset (Both Methods)

		Predicted	
		No	Yes
Actual	No	100	0
	Yes	0	105

Results: Noisy Dataset

- KNN Best: 87.8%
- LA-KNN Best: 88.29%
- Mean k dropped to 3.38

Table 3: Performance Comparison on Noisy Dataset

Method	Parameters	Accuracy	Macro F1
Standard KNN	$k = 9$	87.80%	0.8780
Standard KNN	$k = 15$	87.80%	0.8780
LA-KNN	$m = 5, \alpha = 8.0$	88.29%	0.8829

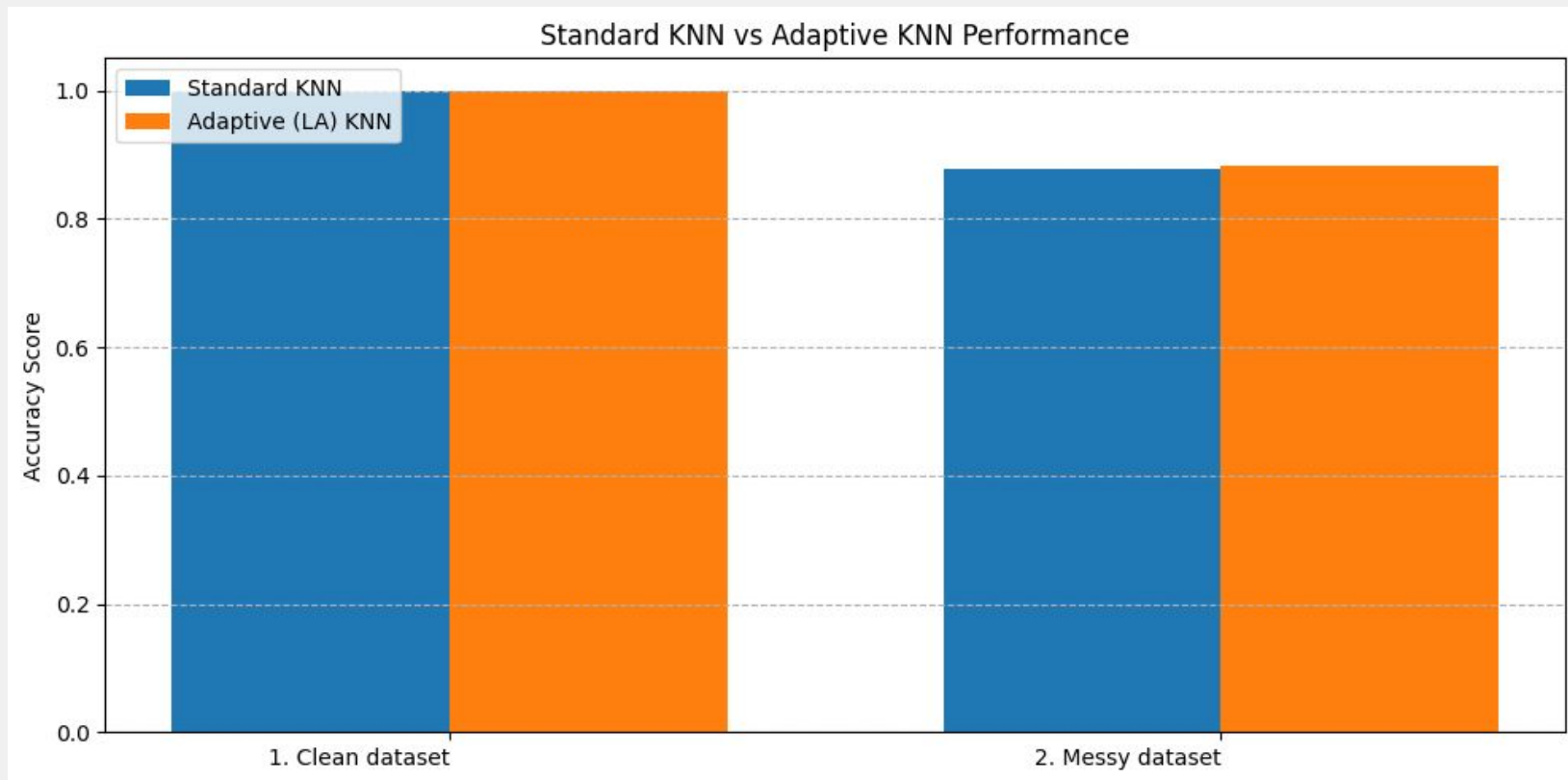
Table 4: Confusion Matrix - Noisy Dataset (Standard KNN, $k = 9$)

		Predicted	
		No	Yes
Actual	No	90	11
	Yes	14	90

Table 5: Confusion Matrix - Noisy Dataset (LA-KNN, $m = 5, \alpha = 8.0$)

		Predicted	
		No	Yes
Actual	No	92	9
	Yes	15	89

Results



LA-KNN Advantages

- Slightly more robust when the dataset is noisy
- Smaller k in scarce regions, Larger k in dense regions
- Reduced influence of neighbors with different labels

Limitations

- Very small performance gains
- Clean dataset
- Improvement came from created noise

Conclusions and Future Work

- Conclusions
 - Same performance as KNN on clean data
 - Slightly better on noisy data
 - Allows doctors to see which patients influenced a prediction
- Future work
 - Evaluate LA-KNN on noisier data
 - Explore different ways to calculate density



Thank you!

References

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